

BSc (Hons.) Data Science and AI Degree Program, IIT Guwahati

Course Content Outline*

(*tentative)

CORE COURSES - Year 1

Trimester I

DA 101 Basic English

Course Outline

The Course will help the learners to develop general proficiency in English in terms of listening, speaking, reading and writing, to gain confidence to use grammatically accepted English for communication, to gain confidence to speak English intelligibly, to learn to use self-study strategies, to use interpersonal communication skills effectively, and become aware of the skills of critical thinking, information transfer and problem solving, develop analytical skills.

DA 102 Data Analysis Basics

Course Outline

Data analysis can be of different types. Descriptive analysis describes or summarizes quantitative data by presenting statistics. Diagnostic analysis is performed to answer why a particular trend is observed in a given data. Predictive analytics uses data to form projections about the future and prescriptive analysis is used for making recommendations. In this course the main focus will be on descriptive analysis and data visualization, however, some aspects of predictive analysis will also be introduced. Spreadsheets are among the most popular tools for data Analysis and visualization. In this course also, data Analysis and visualization will be performed through the use of spreadsheets.

DA 103 Introduction to Statistics

Course Outline

This course provides a foundational exploration of probability theory and random variables, empowering students with essential tools to quantify and navigate uncertainty. From probability basics to mathematical expectations, we delve into topics of random variables, probability distributions, independence, covariance, and correlation. We culminate with the Law of Large Numbers and the Central Limit Theorem, equipping students to confidently analyze data and make informed decisions in a data-driven world.

DA 104 C Programming

Course Outline

C programming language is a general-purpose programming language which is used to develop applications as well for doing system programming. This course is intended to provide foundations for C programming and will also introduce some advanced concepts. Lecture classes will be followed by hands-on so that the learners can develop and execute programs to solve problems.

Trimester II

DA 105 Linear Algebra

Course Outline

In this course you will delve into the beauty of linear algebra, unveiling its power to solve real-world problems. From solving equations to mastering advanced concepts like eigenvalues and matrix calculus, you'll acquire invaluable skills that will act as foundations for advanced topics in data science and machine learning.

DA106 Data Science: An Introduction

Course Outline

In this course, you will get introduced to the exciting landscape of data collection, management, and analysis. From understanding the intricacies of structured and unstructured data to mastering the art of model building and evaluation, you'll acquire the skills and knowledge to unlock the potential of data.

DA 107 Computer System Tools

Course Outline

In this course, you will delve into the fundamentals of computer architecture, operating systems, and system management tools. From understanding the core components of CPUs and GPUs to mastering the intricacies of file systems and user account management, you'll gain the knowledge and skills needed to navigate and optimize both Windows and Unix/Linux environments. The course will uncover the inner workings of computer systems and enable the learners to become proficient in managing and customizing their computing experience.

DA 108 Python Programming

Course Outline

Welcome to the world of Python programming—a gateway to the world of problem-solving through coding! In this course, we'll dive deep into Python's fundamental concepts, from variables and data types to loops, functions, and exception handling. You'll explore the power of data structures like lists, dictionaries, and sets, and venture into object-oriented programming with classes. Whether you're new to coding or seeking to enhance your skills, this course will help you harness the versatility and simplicity of Python, enabling you to unlock your potential as a proficient programmer and problem solver.

Trimester III

DA 109 AI Basics

Course Outline

In this course, you will be introduced to the fascinating realm of intelligent agents, problem-solving, logical reasoning, knowledge representation, and some key topics in machine learning. Through practical examples and case studies, you'll gain insights into the power and potential of AI in various applications.

DA 110 Data Structures

Course Outline

This course is your gateway to understanding essential concepts like arrays, stacks, queues, trees, sorting, searching, graphs, and hashing. By mastering these fundamental building blocks, you'll gain the skills needed to efficiently manage and manipulate data, solving complex problems in programming and beyond.

DA 111 Algorithm Design & Analysis

Course Outline

In this course, you'll delve into the art and science of creating efficient algorithms. From mastering asymptotic notation and complexity analysis to exploring advanced techniques like divide-and-conquer and dynamic programming, you'll gain the skills to solve complex problems systematically. You will gain an understanding of graph algorithms, NP-completeness, and approximation strategies, and equip yourself with the tools to tackle real-world computational challenges.

DA 112 Introduction to R

Course Outline

In this course, you'll embark on a journey into the heart of data analysis and visualization with R. From the very basics of installation and fundamental operations, you'll progress to working with data types, variables, and essential operations. You will implement loops, functions, and data handling techniques, including importing and exporting data files, and explore the art of data visualization and statistical analysis using R's powerful libraries and tools.

CORE COURSES - Year 2

Trimester I

DA 201 Relational Database Management Systems

Course Outline

This course offers a comprehensive exploration of RDBMS fundamentals, from understanding data models and storage structures to mastering SQL queries and transaction management. You'll delve into the intricacies of schema refinement and normal forms, ensuring data integrity and efficiency. You will understand data warehousing and decision support. In the programming laboratory, you'll apply your knowledge to develop real-world database applications, using SQL and front-end tools.

DA 202 Java Programming

Course Outline

This course will introduce you to programming paradigms and offer you a well-rounded perspective on programming approaches. One of the course goals is to enable you to become a versatile programmer,

equipped to tackle a wide range of coding challenges across various paradigms.

DA 203 Optimization

Course Outline

You will embark on a journey to uncover the art and science of solving complex problems efficiently. From understanding the fundamentals of optimization problems and the existence of optimal solutions to exploring a range of algorithms, including the gradient method and simplex algorithm, you'll gain the skills to tackle both unconstrained and constrained optimization, convex optimization and duality, and discover how these principles are applied in real-world scenarios.

DA 204 Basic Econometrics

Course Outline

In this course, we'll journey through the fundamental principles of econometric analysis, from a brief review of random variables to the intricacies of the Classical Linear Regression Model. You'll become proficient in estimating and inferring key parameters, understanding hypothesis testing, and assessing model fit. We'll also delve into the complexities of nonlinear models, dummy variables, and the challenges posed by heteroscedasticity, autocorrelation, and multicollinearity. You'll be equipped to tackle real-world data analysis and interpretation, making meaningful contributions in the field of economics and beyond.

Trimester II

DA 205 Data Mining & Warehousing

Course Outline

In this course, we'll explore the intricacies of uncovering valuable insights from vast datasets. From understanding data analysis basics to delving into data warehousing, we'll journey through similarity search, frequent pattern mining, graph analysis, and data stream mining. You'll be well-equipped to harness the power of data for informed decision-making and uncover hidden patterns in diverse domains.

DA 206 Statistical Inferencing

Course Outline

In this course, we'll explore fundamental principles such as point estimation, interval estimation, hypothesis testing, and advanced topics like multiple linear regression and multivariate analysis. By course end, you'll be well-equipped to draw robust conclusions from data, make informed decisions, and conduct advanced statistical analyses.

DA 207 Signal and Systems

Course Outline

In this course, you'll master the fundamental concepts of signals, systems, and transforms. These foundations are essential for data analysis, machine learning, and AI applications, particularly with unstructured data such as image, video, and audio etc. By course end, you'll have the expertise to process and analyze data for real-world AI solutions.

DA 208 Social Media Tools and Techniques

Course Outline

Social media have transformed the way people interact, influence, and perform business. This course aims to introduce research topics in social media mining and discuss relevant theoretical foundations, methods, and tools. The Lab component of the course will engage students with various state-of-the-art big data analytical frameworks for mining social media content. The class will involve reading papers, presentations, and team projects/assignments.

Trimester III

DA 209 Data Modeling & Visualization

Course Outline

In this course, you'll explore the art of visually representing data. From the fundamentals of data visualization to advanced 2-D and 3-D techniques, you'll gain the skills to convey complex information with clarity and impact. By course end, you'll be equipped to create compelling visual stories from your data, enhancing your data modeling and analysis abilities. Join us on this journey to become a skilled data visualizer.

DA 210 Time Series Analysis & Forecasting

Course Outline

In this course, you'll master the art of analyzing and predicting time-dependent data. From understanding time-series characteristics to spectral analysis and modeling, you'll gain the skills to make insightful forecasts. By course end, you'll be well-equipped to analyze time-series data in various fields, from finance to economics, enabling you to become a skilled time-series analyst and forecaster.

DA 261 Machine Learning Fundamentals

Course Outline

In this course, you'll dive into the foundations of supervised and unsupervised learning, feature selection, dimensionality reduction, and a variety of supervised and unsupervised algorithms. You'll explore real-world applications and case studies, gaining practical experience with tools like Sci-kit Learn and mastering essential machine learning techniques.

DA 262 Recommender Systems

Course Outline

In this course, we'll explore the history, techniques, and evaluation methods behind recommendation systems. You'll delve into collaborative filtering, context-based filtering, hybrid methods, and nearest neighbor approaches, gaining insights into how these systems work. With real-world case studies and a

focus on linguistic and statistical techniques, you'll develop the skills to design effective recommendation systems that consider diverse factors.

CORE COURSES - Year 3

Trimester I

DA 301 Cloud Computing

Course Outline

With the explosion of data and increase in demand for understanding data using machine learning techniques, traditional machine learning solutions are embracing cloud based solutions to meet large scale data and computing needs. This course attempts to augment the present theoretical treatment of machine learning models in the context of advanced cloud based implementations.

DA 302 Deep Learning Essentials

Course Outline

Welcome to "Deep Learning Essentials." In this course, you'll embark on a journey through the exciting world of deep learning. We'll cover the fundamentals of convolutional neural networks, architectural design, loss functions, and sequential modeling. You'll gain insights into visualization and generative models, exploring a wide array of applications. By course end, you'll be well-prepared to navigate the complexities of deep learning and stay updated with the latest trends.

Trimester II

DA 303 Ethics in AI

Course Outline

In this course, you'll explore the ethical considerations and legal aspects surrounding data protection, privacy, and fairness in AI systems. We'll delve into topics like transparency, accountability, and the trade-off between accuracy and interpretability in machine learning. By course end, you'll be equipped to navigate the complex landscape of AI ethics, ensuring responsible and equitable AI development.

Trimester III

DA 304 Soft Skill Enhancement

Course Outline

In this course, you'll hone essential soft skills crucial for success in any profession. From effective communication to teamwork, critical thinking, and professionalism, you'll develop the abilities to excel in your career. We'll focus on adaptability, resilience, time management, and data-driven problem-solving, equipping you to thrive in a dynamic work environment.

CORE COURSES - Year 4

Trimester I

DA 401 Big Data Analytics

Course Outline

In this course, you'll delve into the world of big data, understanding its fundamentals and planning considerations. We'll explore storage concepts, distributed computing frameworks, and stream data processing. Through lectures and hands-on lab exercises, you'll gain practical insights into well-established algorithms, programming assignments, and mini projects.

Trimester II

DA 405 Basics of Reinforcement Learning

Course Outline

In this course, fundamental concepts of Reinforcement Learning (RL) will be taught. Starting with Markov Decision Process (MDP), this course will cover Multi-armed Bandit, policy and value based RL methods. This course will provide you with necessary foundations to solve real-world sequential decision-making problems.

Trimester III

DA 406 Entrepreneurship and Startup

Course Outline

This course imparts foundational knowledge encompassing various facets of creating and managing entrepreneurial ventures across their life cycles. It has been thoughtfully designed to cater to a diverse audience with a multidisciplinary approach. The primary goal of the course is to educate individuals about the critical challenges that entrepreneurs and managers encounter at various stages of a business's journey. It is equally beneficial for aspiring entrepreneurs and decision-makers in established enterprises.

ELECTIVES

Privacy & Information Security

Natural Language Processing

Social Media & Text Analytics

Speech Processing & Recognition

Vision Intelligence

Advanced Applications in AI

Machine Learning Framework

NoSQL

Big Data on Cloud

AI Tools and Applications

Banking & Financial Service Analytics

Business Variables Analysis

Analytics in Securities & Insurance

Analytics & ML in Financial Technologies

Design Thinking

Innovation & Entrepreneurship

Leadership Essentials

Business Research Methods

Agile Development Methods

LANGUAGES, TOOLS, LIBRARIES & REPOSITORIES COVERED

- Statsmodels
 - OpenCV
 - Rest-api
 - Flask
 - Heroku
 - SpaCy
 - mongoDB
 - Shiny
 - Apache HBASE
 - Sqoop
 - Hive
 - Flume
 - PySpark
 - Apache Spark
 - Apache Airflow
 - MapReduce
 - SQL
 - Kafka
 - Pandas
 - Matplotlib
 - NumPy
 - NLTK
 - Seaborn
 - Scikit
 - Kubernetes
 - Docker
 - AWS
 - Excel
 - TensorFlow
 - Keras
 - MySQL
 - Python
 - Tableau
 - PowerBI
 - R Studio
 - GCC (GNU Compiler Collection)
 - Visual Studio Code
 - Ubuntu OS
 - Windows OS
 - C/C++
 - XAMPP
 - JAVA Virtual Machine
 - GNU Octave
 - Jupyter Notebook
 - Anaconda
 - PyTorch
 - CUDA toolkit
 - GitHub
 - Bitbucket
 - Latex Documentation
 - GCP
 - Hugging Face (Gen AI Tool)
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